

Nawal Alisya Binti Mohd Sofiyuddin

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Education

Imperial College London

MEng *Mechanical Engineering with Nuclear Engineering*

September 2023 - June 2027

- Predicted: First Class Honours
- Relevant Modules: Nuclear Thermal Hydraulics, Nuclear Reactor Physics, Aircraft Engine Technology, Machine Dynamics and Vibrations, Mechatronics, Fluid Mechanics, Stress Analysis, Thermodynamics, Heat Transfer, Computing, Nuclear Materials, and Nuclear Fusion.

Experience

MuWave Cooling Loop Design Intern

July 2026 - August 2026

- UK Atomic Energy Authority (UKAEA) FOSTER Programme
- Developed a conceptual water-cooling system for a future gyrotron test facility, defining thermal requirements and evaluating cooling strategies.
- Developed a Python thermal-hydraulics tool for component sizing, pressure-drop analysis and finite-volume modelling of coolant temperature.
- Performed sensitivity analysis on key design parameters and developed the Process Flow Diagram and preliminary system layout.

Amazon Web Services Data Centre Engineering Operations Intern *July 2025 - September 2025*

- Developed an interactive HTML/CSS internal web tool to improve engineer access to complex mechanical and electrical system information.
- Supported preventive maintenance and root-cause analysis of critical mechanical and electrical infrastructure, including chillers, cooling towers, pumps, CRAHs, generators, UPS/PDUs, transformers, switchgear and circuit breakers.
- Analysed HVAC, BMS and EPMS performance data to evaluate cooling demand, system performance and reliability in a high-availability data centre environment.

Malaysia Airlines Engineering and Maintenance Division Internship

July 2022

- Interpreted aircraft maintenance manuals and technical documentation to identify procedures, tooling and components required for Rolls-Royce Trent LP compressor fan blade removal.
- Assisted with Airbus A330 maintenance activities, including cargo fire-extinguisher servicing, avionics isolation and associated circuit-breaker procedures.
- Participated in maintenance of aircraft hydraulic, propulsion and electrical systems, including brake accumulators, engine removal and Integrated Drive Generator (IDG) installation.

Projects

Volumetric Additive Manufacturing 3D Printer

October 2025 - June 2026

- Designed the optomechanical projection system for a Computed Axial Lithography (CAL) volumetric 3D printer using Ansys OpticStudio and experimental validation.
- Identified and quantified barrel distortion and developed a Python predistortion algorithm, reducing experimentally measured TV distortion by approximately 60%.
- Formulated photosensitive resins, characterised optical properties using UV-Vis spectroscopy, and conducted experimental printing trials.
- Designed and implemented a servo-actuated shutter for controlled, repeatable light exposure during printing.

Imperial Formula Racing

Drivetrain Engineer

September 2025 - June 2026

- Designed and manufactured 3D-printed tooling for casting silicone CV boots.

Simulation Engineer

September 2024 - June 2025

- Developed MATLAB vehicle simulations to model internal combustion engine (ICE) fuel consumption and support vehicle performance analysis.

Battery Systems Engineer

September 2023 - June 2024

- Conducted battery module testing and validation using Orion BMS 2, analysing cell voltage and electrical performance to support battery system development.

Pressurised Water Reactor (PWR) Thermal-Hydraulics Model

September 2025

- Developed a Python thermal-hydraulics model for steady-state, single-phase analysis of a PWR coolant subchannel, modelling coolant flow, convective heat transfer and pressure drop.
- Calculated axial coolant and cladding surface temperature distributions using turbulent-flow correlations and a cosine axial power profile.
- Developed a Tkinter GUI for parameter input and results visualisation.

Heat Transfer Simulation Project

March 2025

- Developed a Python thermal model of transient radial heat conduction in a cylindrical resistor with internal heat generation and convective boundary conditions.
- Implemented and compared Explicit Finite Difference and Crank–Nicolson schemes, incorporating centreline symmetry and evaluating numerical stability using the Fourier number.
- Developed an interactive GUI and visualisation framework for parametric analysis, generating temperature profiles, 2D heatmaps and transient animations.

Gibbon Conservation Project

September 2024 - June 2025

- Designed and prototyped a crane with the aim of helping gibbons cross fragmented habitats.
- Performed structural analysis and FEA using Abaqus and SOLIDWORKS, evaluating stress, deformation and load distribution to validate the crane design against hand calculations.
- Designed and manufactured structural components using CAD, milling, turning, spot welding, CNC machining, laser cutting and 3D printing, contributing to assembly and experimental testing of the complete crane.

Awards and Leadership

Project Manager, Volumetric Additive Manufacturing 3D Printer

October 2025 - June 2026

- Led a multidisciplinary engineering team across optics, computation, resin formulation and system integration, coordinating project schedules, technical workstreams, design reviews and project delivery.

Excellence in Design, Make, and Test Project Prize

June 2026

- Awarded for the best Design, Make and Test project in the cohort for the Volumetric Additive Manufacturing 3D Printer.

Engineers and Scientists in Business Fellowship (ESBF) Pitch Competition

March 2026

- Awarded 2nd place for pitching EcoTraveller, a sustainable travel platform concept, following market validation and development of its business model.

Skills

- **Programming:** Python, MATLAB, Simulink, Git, GitHub, and HTML/CSS
- **Engineering Software:** SOLIDWORKS, Abaqus Finite Element Analysis, Ansys OpticStudio, and Ansys Granta Material Selector
- **Languages:** English (Native), Malay (Native), French (Elementary)
- **Certification:** IOSH Working Safely